

L8: Link Clustering Algorithm

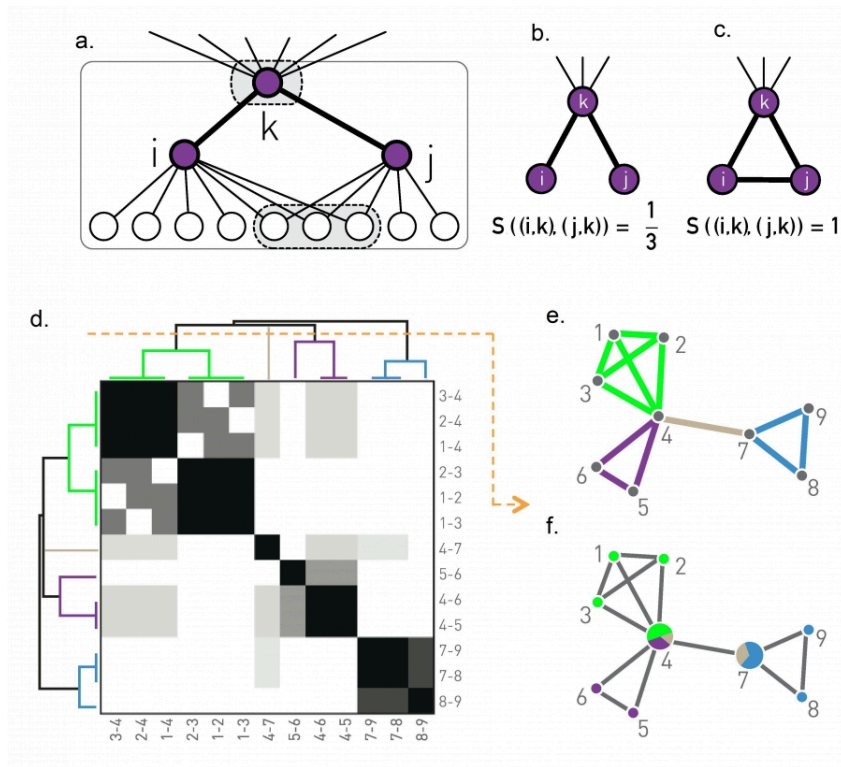


Figure 9.23 from *Network Science* by Albert-László Barabási

Another approach to compute overlapping communities is the link clustering algorithm. It is based on a simple idea: even though a node may belong to multiple communities, each link (representing a specific relation between two

nodes) is usually part of only one community.

For example, a connection between you and a family member is probably only part of your family community, while a connection between you and a co-worker is probably only part of your professional community (unless if your family members are also your co-workers!)

So, instead of trying to compute communities by clustering nodes, we can try to compute communities by clustering links. However, how can we cluster links? We first need a similarity metric to quantify the likelihood that two links belong to the same community.

Recall that if two nodes i and j belong to the same community, we expect that they will have several common neighbors (because a community is a dense subgraph). So the similarity $S((i,k), (j,k))$ of two links (i,k) and (j,k) that attach to a common node k can be evaluated based on the number of common neighbors that nodes i and j have – with the convention that the set of neighbors of a node i includes the node itself. We normalize this number by the total number of distinct neighbors of i and j , so that it is at most equal to 1 (when the two nodes i and j have exactly the same set of neighbors).

In other words, if $N(i)$ is the set of neighbors of node i (including i) and $N(j)$ is the set of neighbors of node j , then the link similarity metric is defined as:

$$S((i,k), (j,k)) = \frac{|N(i) \cap N(j)|}{|N(i) \cup N(j)|}$$

Now that we have a similarity metric for pairs of links, we can follow exactly the same hierarchical clustering process we studied in the previous lesson to create a dendrogram of links, based on their pairwise similarity.

Cutting that dendrogram horizontally gives us different partitions of the set of network links – with each partition corresponding to a community assignment.

These communities, however, even though they are non-overlapping in terms of links, they may be overlapping in terms of nodes.

This is shown in the toy example of part-e: there are four clusters of links, corresponding to three communities of nodes. Note that two of the nodes belong to more than one community.

The computational complexity of the link clustering algorithm depends on two operations; 1) the similarity computation for every pair of links, and 2) the hierarchical clustering computation. The former depends on the maximum degree in the network, which we had derived earlier as $O(n^{2/(\alpha-1)})$ [in L4: Maximum Degree in a Power Law Network] for power-law networks with n nodes and degree exponent α . The latter is $O(m^2)$, where m is the number of edges. For sparse networks, this last term is $O(n^2)$ and it dominates the former term.

Food for Thought

The link clustering algorithm follows a very different approach to perform community detection than all other algorithms we have discussed so far. Can you think of a potential issue with the interpretation of communities in this algorithm?